Climate-associated regions are enriched for ancestry-diagnostic loci, but do not preferentially harbour directional sorting

Draft results summary (Module C, the extrinsic-climate arm). Methods are deferred to Supplementary Materials. Findings are descriptive: climate associations are genotype–environment correlations, and the sorting itself is licensed as selection only against the forthcoming neutral null. The climate-axis identities (which variables load on PC1 vs PC2) are pending (Module C4).

Results

Design, and a controlled ancestry confound. Per-population allele frequencies were associated with two per-population climate principal components (PC1, PC2) using BayPass across eight configurations (two population sets × two PCs × two covariance settings). Because sorting is by definition ancestry fixation, a climate axis correlated with genome-wide ancestry could co-localise with sorting for structural rather than ecological reasons. PC1 is weakly ancestry-correlated (Spearman $\rho = -0.23$), PC2 moderately so ($\rho = +0.57$), driven by a single population, Åland (dropping it lowers the correlation to +0.38; dropping Sielva instead *raises* it, so Sielva exclusion is not the relevant control; Fig. 1). This structure is conditioned out by BayPass's covariance matrix (Ω, estimated on the LD-pruned set) and by the Åland-excluded population set, which we therefore treat as the structure-robust primary.

A size-gated definition, and why it must be replaced. Climate-association outliers are conventionally defined as LD clusters carrying at least N member loci individually significant at BF(dB) $\geq T$. This is an *absolute count*, so eligibility scales with cluster size: on these data a cluster of 9 loci (the background median) would need 56% of its members significant to qualify, whereas a cluster of 64–90 loci (the outlier median) needs only 6–8%; 88% of background clusters contain fewer than 20 loci. Adjusting for $\log(n_{\text{loci}})$ assumes a smooth log-linear size effect and cannot represent this hard, highly non-linear eligibility threshold—it extrapolates the outlier coefficient into a size range in which outliers cannot occur. We therefore replace the binary outlier flag with the per-cluster significance *rate* $p_{\text{sig}} = n_{\text{sig}}/n_{\text{loci}}$, which is size-normalised by construction, retaining $\log(n_{\text{loci}})$ as a covariate, and additionally restrict to clusters for which outlier status is attainable ($n_{\text{loci}} \geq 20$, ≥ 50).

Ancestry-diagnostic loci are robustly enriched in climate-associated regions. Under the size-normalised measure, high-DI loci are enriched in proportion to a cluster's climate-association strength, and the effect *strengthens* as the comparison is restricted to size-comparable clusters (Table 1): odds ratios per +10 percentage points of significance rate rise from 1.35 to 1.50 (PC1) and 1.20 to 1.47 (PC2). Both climate axes show the effect at similar magnitude—the binary criterion had masked PC2 entirely (1.04, n.s. at $n_{\text{loci}} \geq 50$), so the apparent PC1-versus-PC2 dissociation reported under threshold-based definitions is itself an artifact of thresholding.

Directional sorting does not co-localise with climate association. The same reformulation removes the apparent sorting–climate overlap. Under the binary definition, outlier clusters appear 5–8 times more likely to be directionally sorted; that estimate falls to 2.3 (non-significant) once restricted to size-comparable clusters, and under the size-normalised measure it collapses to ≈ 1.0 (Table 1). The point estimate moves *to* unity rather than merely widening, which is the signature of a confound being removed rather than of lost power. A member-level analysis (the fraction of a cluster's member loci that sort—also size-normalised) is likewise null, in agreement.

Interpretation and outlook. The robust climate signal in these data is *static*: regions associated with climate are enriched for ancestry-diagnostic loci, on both climate axes. There is no corresponding *dynamic* signal—directional ancestry sorting is not preferentially located in climate-associated regions once association strength is measured independently of cluster size. We note the limits: the most size-homogeneous stratum contains 451 clusters for the sorting test, so this is "no surviving evidence" rather than a demonstrated absence, and a marginal PC2 effect persists at the intermediate restriction (1.77, $p = 0.035$). On present evidence the extrinsic-climate hypothesis is supported for differentiation but not for the sorting itself. Identifying the PC1/PC2 climate axes (Module C4) and the recombination-matched neutral null remain the decisive next steps; intrinsic incompatibilities are pursued separately via among-population linkage disequilibrium between unlinked diagnostic units.

Table 1: Climate association versus DI-enrichment and directional sorting, primary configuration (Åland-excluded × with-Ω). Odds ratios are per +10 percentage points of significance rate (size-normalised measure) or for the binary outlier flag. Columns restrict to progressively more size-comparable clusters.

Test	Measure	all clusters	$n_{\text{loci}} \geq 20$	$n_{\text{loci}} \geq 50$
DI-enrichment, PC1	rate	1.35^{***}	1.39^{***}	**1.50*****
DI-enrichment, PC2	rate	1.20^{***}	1.32^{***}	**1.47*****
DI-enrichment, PC1 / PC2	binary	1.54 / 1.15	1.59 / 1.11	1.89 / 1.04 n.s.
Sorting overlap, PC1	rate	1.25	1.37	**0.99** n.s.
Sorting overlap, PC2	rate	1.16 n.s.	1.77	**1.03** n.s.
Sorting overlap, PC1 / PC2	binary	5.19 / 8.04	3.50 / 3.12	2.28 / 2.26 n.s.

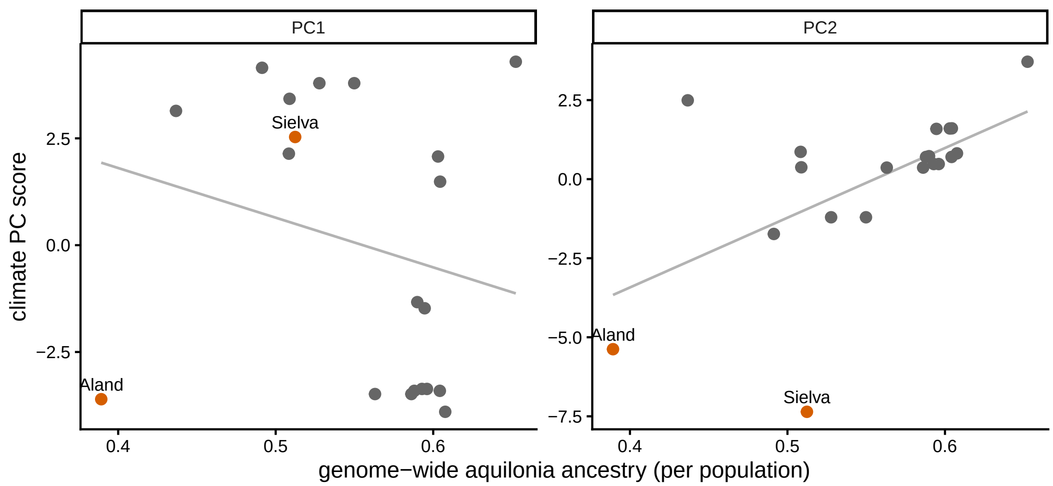

Figure 1: **The climate covariates versus genome-wide ancestry.** Per-population climate PC scores against genome-wide *F. aquilonia* ancestry; Åland and Sielva (orange) highlighted. PC1 is weakly ancestry-correlated; PC2 moderately so, driven by Åland, motivating the Ω and Åland-excluded controls.